



Software product roadmapping in a volatile business environment

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ABSTRACT

Product roadmapping enhances the product development process by enabling early information and long-term decision making about the products in order to deliver the right products to the right markets at the right time. However, relatively little scientific knowledge is available on the application and usefulness of product roadmapping in software product development context. This study develops a framework for software product roadmapping, which is then used to study the critical aspects of the product roadmapping process. The collection of empirical evidence includes both quantitative and qualitative data which sheds further insight into the complexities involved in product roadmapping. Results revealed that organizations view the product roadmap mainly as a tool for strategic decision making as it aims at showing the future directions of the company's products. However, only a few companies appear to have an explicit approach for handling the mechanisms for creating and maintaining such a roadmap. Finally, it is suggested that the strategic importance of product roadmapping is likely to increase in the future and, as a conclusion, a new type of agility is required in order to survive in the turbulent and competitive software business environment.

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1. Introduction

Nowadays, the cost, complexity and rate of technology change are increasing while competition and sources of technology are becoming more global (Phaal et al., 2000a,b). Companies are expected to be more responsive to technological change and to manage their technology assets more strategically because corporate strategies often lack influence over or contact with those who directly manage products (Kappel, 2001). The business environment is characterized by ever-more-demanding customers, increasingly shorter product life cycles and rapidly developing technologies (Groenvelde, 2007). The demand for an overall view of the product and its future releases has become important (Lehtola et al., 2005). Therefore, developing reliable and high quality software products on time and within budget requires a well-coordinated and executed software process (Jiang and Coyner, 2000).

Product roadmapping is an approach to manage a high-level view and to link aspects of business to requirements engineering (RE) (DeGregorio, 2000). It aims at enabling developments in

technology to be mapped and linked to product evolution and market opportunities (Phaal et al., 2003a). Thus, roadmapping aims at enhancing two main purposes in the context of software engineering: how to serve important markets with the right products at the right time and how to improve the cross-functional processes required for the creation of new products (Groenvelde, 2007). Several benefits have been claimed (McCarthy, 2003) to result from the application of roadmapping in organizations. For one, it has been claimed as a simple process for the presentation and cross-functional understanding of complicated issues. It has also been proposed (McCarthy, 2003), for example, to enable a faster and superior assessment of emerging technologies and the contents of products from the learning obtained during the roadmapping process. Also, some difficulties relating to roadmapping have been identified. Yoon et al. (2008) have expressed that many companies, especially medium and small companies, have difficulties in implementing and sustaining roadmapping due to factors such as maintaining the time, cost and effort estimations. A lack of quality input data on markets, competitors and technology, gathered from workshops, complicates the adoption of the method. Furthermore, regular updating of roadmaps can be long and tedious, and may require further workshops and repletion of the roadmapping process (Yoon et al., 2008).

However, currently relatively little scientific knowledge is available on the usefulness and application of product roadmapping in software product development organizations. Research findings

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presented in this paper aim at defining a product roadmapping process and by so doing, increase product roadmapping knowledge for both science and industry. From industry viewpoint, this study aims at increasing the product roadmapping knowledge in order to assist companies in developing their own product roadmapping activities. From scientific perspective, the study aims at increasing the body of knowledge in the area of product roadmapping while also identifying future research and development work opportunities on the most critical aspects of the product roadmapping process. The research is structured and presented by using a research framework for software product roadmapping, which is created based on the current literature. The research framework provides the lenses that guide the empirical analysis. These lenses are defined as follows: defining the main stakeholders and their roles during the process, organizing the product roadmapping process, establishing the main benefits and challenges faced during the process, and identifying the most critical phases of the process. The research provides empirical evidence from altogether nine interviews and a questionnaire obtained from 34 different companies.

This article is structured as follows: in section two, product roadmapping is defined based on the current literature including an overview of roadmapping, stakeholders of the product roadmapping process and product roadmapping phases. In section two, a software product roadmapping framework is also presented based on the existing literature including the central elements of roadmapping, and the perceived impacts of software product roadmapping and its main phases. Thereafter, in section three, the research design of this study is presented, which includes the research methods and goals as well as the research context and data collection methods. In section four, the research evidence from the empirical study is presented and aligned with the presented research framework. In section five, the research results are discussed and, finally, in section six, conclusions are drawn.

2. Related work

In this section, we will outline the current knowledge on product roadmapping based on the existing literature and research. Additionally, in this section, a research framework for software product roadmapping is proposed based on the existing literature. Firstly, the terminology of roadmapping is introduced (Section 2.1). Then, stakeholders (Section 2.2) of product roadmapping, and the proposed main phases (Section 2.3) of product roadmapping are specified. Finally, a research framework for software product roadmapping (Section 2.4) is presented.

2.1. Overview

A roadmap is a layout of existing routes or paths, which is used to decide among alternative directions towards a desired destination (Kostoff and Schaller, 2001). It is said by DeGregorio (2000) that a roadmap is a visualization of a forecast, which can address a number of key areas, such as technology, capability, platform, system, environment, threat, and business opportunity. According to Kappel (2001), roadmaps are also forecasts of what is possible or likely to happen, and plans that express a course of action. Furthermore, roadmaps are intended to be living documents, to be reviewed and updated over time in order to remain useful (Albright, 2003). Roadmapping describes the process of creating and revising roadmaps (Kostoff and Schaller, 2001). It is a strategic planning and forecasting process with long-lasting future activities (Kappel, 2001). Roadmapping is also claimed (Li and Kameoka, 2003) to be a decision-making and design process. On the other hand, Strauss et al. (1998) describe roadmapping as a management activity that links customer/market needs and opportunities, product quality

and competitive positioning as well as corporate capabilities such as business and technology value chain attributes.

Roadmapping can be seen similar to strategic planning. However, strategic planning does not mean necessarily strategic thinking, and thus it is claimed by Mintzberg (1994) that the most successful strategies are visions, not plans. When the difference between planning and strategic thinking is understood, the company can get back to the strategy making process: capturing what the manager learns from all sources. Furthermore, according to Mintzberg (1994) strategic planning is about breaking down a goal or a set of intentions into steps, formalizing those steps to be implemented almost automatically, and then articulating the anticipated consequences or results of each step. On the contrary, strategic thinking is about synthesis and involving intuition and creativity. The strategic thinking results in an integrated perspective of the company, but not too precisely articulated vision of direction (Mintzberg, 1994). For example, Beeton et al. (2008) have classified roadmapping strategies into two categories. Either the roadmapping may be exploratory, i.e. surveying future possibilities, or goal-oriented, i.e. defining strategies to realise clearly defined future targets.

Roadmaps can be expressed in various forms, types or with different taxonomies (Kameoka et al., 2003). For example, according to Kappel (2001) roadmaps can be categorized into science or technology roadmaps, industry roadmaps, product-technology roadmaps, and product roadmaps. Kappel (2001) describes the interrelationships between these four types of roadmaps as follows. The purpose of the science or technology roadmaps is to set industry targets and understand the future by identifying trends and making accurate forecasts. The industry roadmap is a combination of technology forecasts and industrial matters. Thus, the purpose of the industry roadmap is to set industry expectations that express for example a technical thrust and a competitive environment. The product-technology roadmap is a combination of specific product plans with technology trends and marketplace. The purpose of the product-technology roadmaps is to align product and technology generations together. On the other hand, the product roadmaps schedule product introductions, such as a direction and schedule for product evolution, which are then communicated with customer and internal audience (Kappel, 2001).

The taxonomy according to Kappel (2001) is presented in a chart, in which the horizontal axis describes the roadmapping purpose on the industry or company level, and the vertical axis describes the content emphasis of the roadmap either on specific trends or on positioning within an industry. In addition, Phaal et al. (2005) claim that roadmaps commonly take a form of a multi-layered time based chart that includes different layers of knowledge relating to purposes, deliveries, and resources. On the contrary, Vähäniitty et al. (2009) state that the product roadmap consists of five layers with the four topmost depicting activities and the bottom layer illustrating the estimate of human resource requirements. The activities are such as performing services, preparing releases, developing product components and platforms. According to Albright (2003) and Groenvelde (2007) the multi-layered roadmap can be constructed from a market-pull or from a technology-push perspective. From the market-pull point of view, a roadmap should begin with defining the most important requirements of the marketplace and customers. This strategy includes defining product development in the process of time and defining the required technologies for these products. From the technology-push viewpoint, a roadmap should begin with defining the key or new technologies and their market needs. This strategy describes how technology is going to affect the functionality of the product (Albright and Kappel, 2003; Groenvelde, 2007). Even though roadmaps may take various forms or taxonomies, they all should answer a common set of “why-what-how-when” questions that generally relate to markets,

products, and technologies (Phaal et al., 2005). However, as Phaah et al. (2004) emphasize, the form of the roadmap should be tailored to the specific needs of the company and its business context.

A product roadmap provides a forecast of product family evolution over time and views the whole platform or relationships between the products in a platform (Albright and Kappel, 2003). Typically, product roadmaps are developed, reviewed and improved iteratively. Often, this is achieved through close human interaction such as face-to-face meetings and workshops with relevant stakeholders (Phaal et al., 2005). Typically, product roadmaps are owned by the business owner of the product, who is also responsible for gathering all relevant stakeholders to obtain the needed information for the roadmaps. Product roadmaps typically cover a scope of two to three years, during which time they are to be frequently revised in order to ensure the currency of the documentation (Tabrizi and Walleigh, 1997; Lehtola et al., 2005). Documentation of features is an important part of product roadmapping, since accurate documentation ensures that the roadmaps can be read, analysed, redrawn, and validated (Nuseibeh and Easterbrook, 2000). The information in the product roadmaps incrementally describes how the product and its business environment changes yearly. The fields in the product roadmap define the high-level functionality of the product and target customer group. The high-level product functionality is a description of forthcoming releases with basic mandatory information. The mandatory information includes the release goal and high-level features for each release. In addition, release time, localization, platforms, and dropped topics, e.g. features that are not supported in the subsequent versions of the product, are recorded. Additionally, the fields include information about positioning, market arguments, and the geographical focus of the product for every year. This information is represented with a few bullet points for each issue (Lehtola et al., 2005).

Product roadmapping pertains to the software engineering (SE) process as it is a method for planning and defining software product requirements based on the market and needs of stakeholders. Software systems requirements engineering (RE) is defined as a process of discovering a system's purpose, by identifying stakeholders and their needs, and documenting these in a form that is adequate to analysis, communication, and subsequent implementation (Nuseibeh and Easterbrook, 2000). Product roadmapping is also closely linked to RE. For example, according to Jantunen and Smolander (2006) product roadmapping deals with the different releases of each product. In a product roadmap, the product is represented as product releases containing several product features. According to Wieggers (2003), a product feature is a set of logically related requirements that provide a capability to the user and enable the satisfaction of a business objective. Instead, a requirement can also be a condition, which a product must meet to satisfy such a need or objective. In other words, a requirement is a property that a product must have to provide value to a stakeholder. However, product roadmapping differs from RE in that it is a process with long-lasting future activities, and in roadmapping, high-level features are presented within a timeline and scheduled for different releases. On the contrary, in RE, the features are analysed in more detail and defined as to what they mean from the perspective of the product development project. Also, Lehtola et al. (2005) have pointed out that targets for the release roadmapping and the product roadmapping are quite the same but the product roadmapping is more high-level. The main objective of the release roadmapping is to inform stakeholders about scheduled future releases. Instead, the objective of the product roadmapping is to help product managers create and maintain release roadmaps, manage situations where the same technical product is included in several products, and R&D to identify the needs for research

projects (Lehtola et al., 2005). Furthermore, product roadmapping can be seen as a part of broader perspective to software product management. According to a reference framework for software product management (van de Weerd et al., 2006), the product management consists of portfolio management, product roadmapping, requirements management, and release planning. In that framework, product roadmapping includes theme identification, core asset identification, and roadmap construction. Instead, requirement management includes activities of gathering, identifying and organizing requirements, and release planning includes activities such as requirements prioritisation, selection, definition, validation, scope change management and launch preparation.

2.2. Product roadmapping stakeholders

It has been suggested (Tabrizi and Walleigh, 1997) that product roadmaps are created by senior management who are also responsible for updating the roadmaps. However, it has been argued (Li and Kameoka, 2003; Lehtola et al., 2005) that the roadmapping team more usually consists of various stakeholders from different functions of the organization or even from different organizations. The team shares information and perspectives to make decisions that are then presented in a roadmap (Lehtola et al., 2005). The study of Lehtola et al. (2005) supports the latter approach, though it does not propose who the participants are, and what their role in the roadmapping process is. Nevertheless, the research of Lehtola et al. (2005) indicates that the most important stakeholder groups to which the contents of the roadmaps should be communicated and with which they should be negotiated were product management, sales and channel partners, and customers. The product developers were not seen as an important stakeholder group in the roadmapping process, yet they were considered important in estimating the costs of future requirements.

According to McCarthy (2003), only the roadmapping team participates in the roadmapping process while support from management is needed regarding personnel and budget investments. The team should be formed at the beginning of the roadmapping process, including the research and development (R&D) and technology management personnel, members from business development, representatives from finance, and core staff members from the other functions. The first task of the team is to establish a common understanding of the process and the terminology to be used. After that, the team should begin to develop a detailed analysis of the process, and to decide factors and metrics required for the process evaluation. The roadmapping team is also responsible for analysing the required technologies as well as implementing and reviewing the roadmaps.

On the other hand, Groenveld (2007) proposes that the roadmapping process should be started with a small roadmapping team in which the marketing, product management, research, development, and engineering teams participate. Later, the team looks for a leader who should become the owner of the drafted roadmaps. The owner is responsible for the maintenance of the roadmaps and for the initiation of appropriate updating actions as well as providing additional information when needed. The roadmapping team guides the process and organizes workshops to ensure integral involvement of the organization and input by the organization. The outcome of the workshops is used to prepare draft roadmaps, or parts of them.

According to Phaah et al. (2000a,b, 2003a), a multifunctional team is needed in the roadmapping process in order to provide multiple perspectives such as commercial, technical, research, development, manufacturing, marketing, and finance. In addition, Phaah et al. (2003a) believe that both the business owner and the process owner should participate in the roadmapping process. The owners should be involved in the planning phase and, thereafter, through-

Table 1

Summary of the roadmapping process stakeholders.

Stakeholders	Reference							
	Tabrizi and Walleigh (1997)	Lehtola et al. (2005)	McCarthy (2003)	Groenveld (2007)	Phaal et al. (2000a,b), Phaal et al. (2003a)	Albright (2002)	Rautiainen et al. (2003)	van de Weerd et al. (2010)
Senior management	X						X	
Product management		X		X		X	X	X
Sales and channel partners		X						X
Customers		X					X	X
Representatives of R&D			X	X	X	X	X	X
Technology management			X					
Representatives of business development			X					
Representatives of finance			X		X			
Representatives of marketing				X	X	X		X
Representatives of engineering				X				
Representatives of manufacturing					X	X		
Representatives of services						X	X	X
Representatives of support								X

out the roadmapping process. The business owner is proposed to be responsible for the business outcome of the process, while the process owner is responsible for the implementation of the roadmapping (Phaal et al., 2003a; Wells et al., 2004). The owners are also responsible for selecting the persons to the roadmapping team, solving issues regarding the application, and having knowledge about the roadmapping domain (Phaal et al., 2003a). In addition, a facilitator is proposed for managing and facilitating the roadmapping process (Phaal et al., 2003a; Wells et al., 2004). During the different phases of the roadmapping, a facilitator can support and guide the roadmapping team (Albright, 2002). Moreover, according to Albright and Kappel (2003), facilitators have active roles in appropriately scoping the roadmap, forming the team, setting up a work plan, and assessing individuals with their tasks in the larger effort. The facilitator should also challenge assumptions and force rigour into the roadmap (Albright and Kappel, 2003).

Albright (2002) also suggests that the roadmapping process is best performed as a cross-functional team led by an experienced facilitator. Through the whole process, the facilitator steers the team towards a realistic plan. The cross-functional team includes many functions that contribute to the success of a product line or business: central and regional marketing, product management, R&D, manufacturing, services, etc. The purpose of the roadmapping team is to lay out a possible future or multiple futures, set objectives, and define a plan to achieve the objectives, as well as make sure that the required capabilities and technologies are available at the right times (Albright, 2002). Furthermore, Jantunen and Smolander (2006) have identified three types of roles that need to be present in a roadmapping process: contributor, controller, and distributor. Contributor is the one that brings valuable information to a roadmapping context. Controller ensures that roadmapping is done systematically, and distributor absorbs information at a roadmapping context and disseminates the results of roadmapping to those who need to act upon it. Contributor has similar tasks as the previously mentioned member of a roadmapping team, and controller acts in a quite similar role as a facilitator.

van de Weerd et al. (2006, 2010) have made a difference between internal and external stakeholder groups. Their definition of the product management stakeholders is adopted from (Gorchels, 2000; Lehtola et al., 2005). The internal stakeholders are described as follows: product management, company board, research and innovation, services, development, support, and sales and marketing. The external stakeholders are defined as follows: market (i.e. potential customers, competitors, and analysts), partners (i.e. implementation, development, and distribution partners), and customers.

Clearly, in literature there are differing views on how widely and in what role relevant stakeholders should participate in the roadmapping process. Just as clearly, the different views are largely based on opinions or views and not on empirical research about how the roadmapping process is actually carried out in practice and what kinds of problems and challenges have been met in practice. Table 1 provides a summary of the stakeholders of the roadmapping process presented by different authors.

2.3. Product roadmapping process

The roadmapping process focuses on sharing perspectives, involving interaction between people, leading to communication, new understanding, insights, creativity, and learning (Phaal et al., 2005). However, due to the different markets and cultures of companies the roadmapping process varies between companies (Phaal et al., 2004; Groenveld, 2007). According to Phaal et al. (2004), finding a suitable roadmapping process depends on many factors such as the level of available resources (e.g., people and time), issues to be addressed (e.g., purpose and scope), information available (e.g., market and technology), as well as other existing processes and management methods (e.g., new product development, project management and market research).

Actually, only a few process descriptions focusing on product roadmapping seem to have been proposed in the literature. Albright and Kappel (2003) define the roadmapping process to include stages of initiation, maintenance, and restart, based on an organisation's specific needs. Phaal et al. (2003a) identify three roadmapping phases: planning, facilitated roadmapping workshop(s), and roll-out. A fairly similar three-phase product roadmapping process has been proposed also by Lehtola et al. (2005) including preparation, approval, and communication. On the contrary, van de Weerd et al. (2010) define product roadmapping process to consist of theme identification, core asset identification, and roadmap construction. Alternately, McCarthy (2003) adds two more phases to the roadmapping process that consists of team formation, focus, technology or workflow analysis, implementation, and review. Moreover, Vähäniitty et al. (2002) propose a four-step model especially for creating and updating product roadmaps. The following steps are suggested: (1) Define the strategic mission and vision of the company, and outline the product vision. (2) Scan the environment by identifying major trends. (3) Revise and distil the product vision as product roadmaps. (4) Estimate the product life cycle and evaluate the mix of development efforts planned. The steps in the model should be performed periodically to adjust the roadmap to new information

Table 2
Summary of roadmapping processes.

Name	Goal	Main phases	Reference
Roadmapping life cycle	Goal is to define and communicate product and technology strategy along with a longer, smarter view of the future	Initiation Maintenance Restarts	Albright and Kappel (2003)
T-Plan: fast-start technology roadmapping (TRM)	Goal is to bring together key stakeholders and experts to capture, share and structure knowledge about the issue being addressed, to identify strategic issues and to plan the way forward	Planning Facilitated workshop(s) Roll-out	Phaal et al. (2003a)
Product roadmapping (PRM)	Goal is to help (1) product managers create and maintain release roadmaps, (2) managing situations where the same technical product is included in several products, and (3) R&D to identify the needs for research projects	Preparation Approval Communication	Lehtola et al. (2005)
Release roadmapping (RRM)	Goal is to inform stakeholders about scheduled future releases to help R&D, for example, plan their skills development and act as a trigger for early feature development, or marketing, to plan their future activities	Data collection Feature prioritisation Release planning Release roadmap validation	Lehtola et al. (2005)
Roadmapping process	Goal is to improve internal processes which may need improvement to increase R&D productivity or to upgrade a step in the drug discovery process that has fallen behind “industry standards”	Team formation Focus Technology/workflow analysis Implementation Review	McCarthy (2003)
Four-step model for creating and updating product roadmaps	Goal is to define and concretise the company's plans for technology and product development	Define strategic mission and vision, and outline product vision Scan the environment Revise and distil the product vision as product roadmaps Estimate product life cycle and evaluate the mix of development efforts planned	Vähäniitty et al. (2002)
Product roadmapping	Goal is to handle the development of the product roadmap, in which the future releases are planned based on themes and core assets	Theme identification Core asset identification Roadmap construction	van de Weerd et al. (2010)

and changing market situations. Smaller updates to the roadmaps are suggested to ensure up-to-date information (Vähäniitty et al., 2002).

However, unlike the others, Lehtola et al. (2005) separate the release roadmapping process as distinct from product roadmapping. It is proposed to consist of data collection, feature prioritisation, release planning, and release roadmap validation. Similarly, van de Weerd et al. (2010) separate the requirements management and release planning as distinct from product roadmapping. The requirements management consists of gathering, identifying, and organizing requirements. The release planning includes requirements prioritization, requirements selection, release definition, release validation, launch preparation, and scope change management. In Table 2, a summary of the existing roadmapping processes is presented including their main focus areas, i.e. goals and main phases.

2.3.1. Capturing features into roadmaps

According to van de Weerd et al. (2010), the product roadmapping starts with identifying themes and core assets for the future releases. Instead, according to Vähäniitty et al. (2002), the product roadmapping process starts with defining or revising, and then analysing the strategic mission and vision of the company. The

purpose is to clarify and communicate the company's area of business. This is because all companies should have a sufficiently clear idea of their purpose and desired future to be written down before their operations are planned in more detail. The company's mission and vision acts as a guideline for shaping the product vision and choosing between strategic alternatives. Thereafter, major trends in the business environment, such as potential customers, competitors, the industry and developments in relevant technology, are observed and identified. The purpose is to create an understanding of the desired focus and position of the company and its products, as well as examine and guide the selection of technology (Vähäniitty et al., 2002).

According to Nuseibeh and Easterbrook (2000), capturing or eliciting of features is regarded as the first step in the requirements engineering process. When capturing features, gathered information often needs to be interpreted, analysed, modelled and validated to be sure that a sufficiently complete set of features of a product have been collected. Thus, capturing features closely relates to other roadmapping activities.

One of the most important activities in the capturing features phase is to find the problems that need to be solved, and hence identifying the product boundaries. The boundaries define, on a high level, where the final delivered product will fit, e.g. with

which target markets and which potential customers. Identifying and agreeing on the product's boundaries affect all the following feature-capturing activities. Therefore, the identification of stakeholders, user classes, goals, tasks, scenarios, etc. all depends on how boundaries are chosen. The identified stakeholders are persons or companies who stand to gain or lose from the success or failure of the product. Usually, the stakeholders include customers or clients, developers, and users, for instance. In addition, goals should be captured early in the product roadmapping process. The goals should denote the objectives a product must meet, and focus on the needs of the stakeholders (Nuseibeh and Easterbrook, 2000). Thereafter, the product features can be identified and gathered by communicating with all stakeholders (Parviainen et al., 2003).

There are several methods, which can be used during the feature capturing process. These methods include contextual inquiry, observation, prototyping, and scenarios (Parviainen et al., 2003). However, based on the literature, features are captured into product roadmaps by using, for example, group elicitation techniques. For instance, Nuseibeh and Easterbrook (2000) suggest that these techniques aim at improving the stakeholder agreement and buy-in, while exploiting the team dynamics to elicit a richer understanding of needs. According to Phaal et al. (2003a), the group elicitation technique includes a workshop or series of workshops which bring together a range of expertise, supporting the rapid capture, structuring and sharing of knowledge, together with simulating and brainstorming participation. During the workshop, all ideas for features in a product are collected in a roadmap template, which is also called product backlog or product feature document. The document provides a systematic way of collecting feature suggestions continuously from all participants and stakeholders (Phaal et al., 2003a; Rautiainen et al., 2003). There are also other techniques used in capturing features into roadmaps, such as market research, interviews, surveys and analysis (Phaal et al., 2003a). This is because product roadmaps require a good understanding of the markets and application in order to define the products in terms of customer needs (Groenvelde, 2007).

In the capturing features phase, companies can collect features from several preferred sources for the product. Parviainen et al. (2003) propose the following sources as inputs for this phase: business requirements, customer requirements, user requirements, constraints, in-house ideas and standards. In addition to these perspectives, Albright and Kappel (2003) point out that the roadmapping team can define a market section which includes an analysis of competitors, a market research, and a product evaluation.

2.3.2. Analysing features

When the product features have been collected, the roadmapping team should begin to analyse the collected features. There are several methods that can be used during the feature analysis, for example knowledge-based critiquing (Fickas and Nagarajan, 1988), and feature-oriented approach to model requirement dependencies (Zhang et al., 2005). The purpose of the analysis is to remove uncertainty, identify and resolve conflicts as well as to analyse the feasibility of the gathered features, and make a resource and cost estimation (Soffer et al., 2005). Also, the purpose is to reveal dependencies between the requirements. Thus, the team should decide the methods required to evaluate the features.

Other issues to be considered in feature analysis are predictability of outcome, internal competencies in the organization, and opportunities for technology improvement (McCarthy, 2003). Also, the company's internal factors, such as human and financial resources, competencies and infrastructure, should be taken into account when analysing the features (Vähäniitty et al., 2002). Furthermore, there might be some factors restricting or improving the

product features. For example, in the development of the mobile phone, the amount of the memory affects the features that are going to be included in the phone. This is because the memory might not be sufficient to implement all the presented features. Therefore, in the analysing phase, it should be analysed which features can be included in the phone and which have to be excluded. On the other hand, new technologies and development methods, unavailable earlier, can improve the product development; thus, these possibilities should also be analysed.

Thereafter, the gathered features and capabilities should be mapped into groups (Phaal et al., 2003a). According to Albright (2002), the features should be grouped by the product drivers that the features most strongly affect. Instead, according to Phaal et al. (2003b), the features should be first grouped, and then the groups should be arranged in terms of impact on the market and business drivers defined in the earlier phase.

The feature grouping is followed by feedback and discussion to identify synergies and gaps. The purpose of this practice is to ensure that all layers of the roadmap have been considered (Phaal et al., 2003a). If gaps are found, the team should take actions to close the gaps, e.g. by filling them with new product features. The gaps may also include a key technology that must be included in the product to meet a high-priority customer and market need. Hence, it should be estimated whether to develop or acquire the needed key technology (Albright and Kappel, 2003). Also, in this phase, the impact on market and the business drivers of the gathered product features are assessed, and alternative product strategies are considered (Phaal et al., 2003b; Holmes et al., 2004). Based on the feature analysis, the product vision is revised and captured as product roadmaps (Vähäniitty et al., 2002).

2.3.3. Prioritising features

Prioritising features is difficult and time-consuming, since features are related to each other. Therefore, it is complicated to schedule features based on priority only. Hence, at this point interdependencies between features should be explored, identified, and managed (Carlshamre et al., 2001). In product roadmapping, the product features should be prioritised so that the most important features are implemented first and the less important features are left until later, and the least important features are most likely omitted if the schedule or budget is insufficient (Greer and Ruhe, 2004).

According to Albright and Kappel (2003), the most important goal of roadmapping is to identify and focus strategy and product development on the few most important elements for success. Therefore, the roadmapping team should try to define the two or three most important drivers, elements or issues. That is, identifying the highest priorities. Albright (2002) points out that to achieve the main objectives, the team can define an action plan of a roadmap. The action plan identifies the highest priority features, and leads the team to schedule, budget, and staff them to accomplish the goals. With the action plan, the team can make sure that all feature gaps are closed.

There are different methods and tools for use during prioritisation. For instance, features can be prioritised by using informal and formal prioritisation methods. From the informal prioritisation method viewpoint, for example, Blotner (2004) suggests that initial feature prioritisation is done by the roadmapping team only using the identity information. This means that each feature is presented and input from all team members is gathered. Then the team attempts to agree on a spot for the product feature in the feature priority list. If consensus cannot be reached, the project manager either makes the final decision concerning the prioritisation or gathers enough information for the team to come to a consensus. Also, according to Phaal et al. (2003a), the roadmapping team conducts the prioritisation of the product features. The priori-

tisation of features is based on preparing an outline communication roadmap so that the priorities can be identified through feedback and discussion in a workshop session.

There are also theories that support the fact that formal feature prioritisation methods are used during product roadmapping. For instance, Phaal et al. (2003b) refer to Quality Function Deployment (QFD¹) (e.g. Griffin and Hauser, 1993) as an often-used method for supporting product design in product roadmapping. Also, according to Groenvelde (2007), the roadmapping process is quite frequently supported by methods such as QFD, because it is a customer-oriented approach that guides the roadmapping team at the beginning of the product roadmapping process. Typically, QFD is used because it is similar to roadmapping; they both require multidisciplinary communication and decision-making. QFD also helps to focus on the market requirements and translate these requirements into appropriate product characteristics, which facilitate feature prioritisation and, in consequence, product roadmapping. Groenvelde (2007) also points out that there is one essential difference since roadmapping implies both current and future new features whereas QFD is generally restricted to available features. According to McCarthy (2003), the simplest way to use QFD is to develop a matrix in which the product needs are listed on the left side of the matrix and along the top are listed the features used to address these needs. The degree of alignment is rated pertaining to how well the feature meets the needs.

After prioritising the features, when the relative priority of each feature is established, product roadmap construction should start from defining the major and minor release cycles (Vähäniitty et al., 2002; Wiegers, 2003). The construction should be continued with defining the business features and expectations for the upcoming releases. When business features and their objectives are included in the feature repository and their history is traced, the rationale behind the roadmap evolution will become visible (Vähäniitty et al., 2002). With feature prioritisation, the construction of the product can be planned to provide the highest value at the lowest cost. However, even the low priority features should be documented, because their priority might change later and knowing them will help developers to plan future enhancements (Wiegers, 2003). Prioritisation is also important if features are dropped in case they cannot be finished in time for the release (Rautiainen et al., 2002).

2.3.4. Roadmap validation and agreement

In the roadmap validation and agreement phase, the planned product life cycle is estimated and the mix of development efforts is evaluated. The purpose is to check the financial rationale and assess whether the planned development is compliant with the product and company vision (Vähäniitty et al., 2002). Also, missing features and inconsistencies are discovered (Grynberg and Goldin, 2003). At the same time, the content of the roadmap and its key messages are considered, and the gathered data is validated with internal expert information (Wells et al., 2004). Thus, suggested by McCarthy (2003), the roadmapping team should review the product roadmaps to determine whether the goals of the roadmapping effort have been met. If modifications to the roadmaps are required, the roadmapping team should define a revised action plan.

There are different variations how product roadmaps can be validated. For instance, according to Phaal et al. (2003a), the roadmap validation happens by ensuring that all layers of the roadmap are considered and all necessary information is included. Instead, according to Nuseibeh and Easterbrook (2000), roadmap validation

is conducted by identifying the most important goals of each participant and then ensuring that these goals are met in the roadmap.

Finally, all product releases within the scope of two to three years and their content, i.e. features, are agreed and possible disagreements among stakeholders are resolved. This phase calls for effective communication on the product releases and features between different stakeholders, especially if the stakeholders have different goals (Nuseibeh and Easterbrook, 2000; Grynberg and Goldin, 2003). Thereafter, when the strategic issues are identified, discussed, and actions agreed the product roadmapping process can be taken forward (Phaal et al., 2003a).

2.3.5. Change management of the roadmap

Change management is part of product development but it also affects roadmapping, since product roadmaps should evolve as the environment, in which the product operates, changes and stakeholder needs change. Thus, managing the changes is a fundamental activity in the product roadmapping process. In product roadmapping, changes are managed by using tools for configuration management and version control, and exploiting traceability links to monitor and control the impact of changes in different parts of the roadmap documentation (Nuseibeh and Easterbrook, 2000). Furthermore, Richey and Grinnell (2004) have suggested that maintaining the roadmaps could be supported with building a composite roadmap digitally. According to them, it is a fast and simple technique and it allows the owners of each portion of the roadmap to maintain control of edits and changes.

Typically, changes to product roadmaps include adding or deleting product features, and fixing errors. Features are added because stakeholder needs change or because they were missed in the initial analysis (Nuseibeh and Easterbrook, 2000). When new features are added to the roadmap, it means that some of the other features must be excluded (Rautiainen et al., 2003). Usually, features are deleted during development to prevent cost and schedule overruns (Nuseibeh and Easterbrook, 2000). Fixing errors and improvement suggestions are included in the product roadmap and planned into future product releases. These re-prioritisations of the product features are the responsibility of the product roadmapping team. Further, the priorities are reformed in different sessions with internal experts and selected customers and partners (Rautiainen et al., 2003).

The requirement change management process defined by Pozgaj et al. (2003) constitutes the following phases: feature change identification, analysis of the change, definition of the change impact, definition of the change actions, decisions, and implementation of the feature change. The roadmapping team members are responsible for continuously tracking and supervising features in order to discover features' changes. When the indications for feature change occur, the features should be carefully analysed and the feature change identified. After the change identification, all consequences of the change must be analysed (Pozgaj et al., 2003). Each proposed change should be evaluated in terms of the existing features and architecture so that the trade-off between the cost and benefit of making a change can be assessed (Nuseibeh and Easterbrook, 2000). The feature change analysis should focus on the impact of the feature change on other features and on the definition of the features' change actions (Pozgaj et al., 2003). The purpose of the feature impact analysis is to identify what to modify to accomplish a change, or to identify the potential consequence of a change (Arnold and Bohner, 1993). Traceability links help to scope a possible impact of change and to define which parts are related to which other parts according to specific relationships (Arnold and Bohner, 1993; Nuseibeh and Easterbrook, 2000). The impact analysis is important, since product features are frequently interdependent. Thus, a small change could create a major impact because of the many ripple effects (Ebert and Smouts, 2003). The feature change

¹ Drs. Yoji Akao and Shigeru Mizuno originally developed QFD in the early 1960s. They also co-founded the Quality Function Deployment Institute. <http://www.qfdi.org/>.

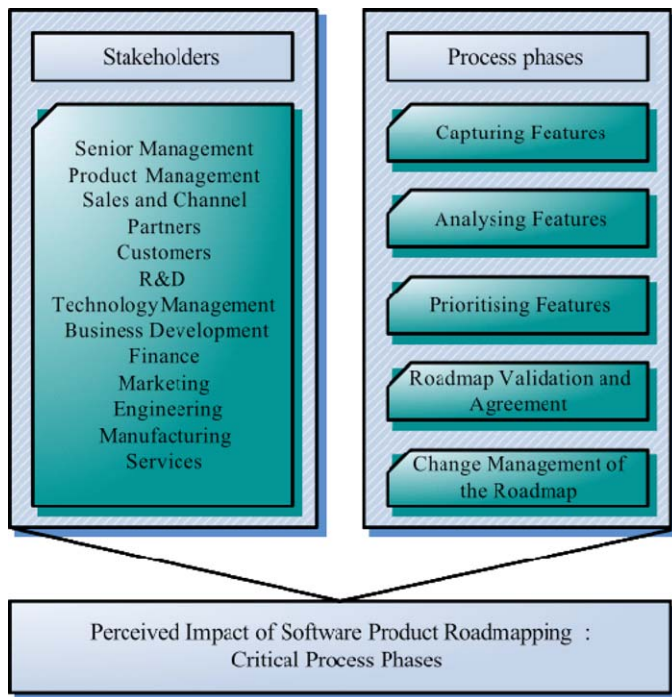


Fig. 1. Research framework for software product roadmapping.

actions should be defined for all development areas that are under the impact of feature change to enable immediate response to the feature change. The purpose is to provide sufficient information for the decision about the realization of feature change (Pozgaj et al., 2003).

Finally, a Change Control Board (CCB) makes all the decisions concerning the change. The CCB can be the same as the roadmapping team or the company can have a separate group of people forming the CCB. On the other hand, the CCB can also be a part of the roadmapping team. In that case, only the most important representatives, e.g. the owner, participate in the decision-making. The CCB decides on each feature change request according to the information provided by the feature change analysis to implement or to refuse the feature change (Pozgaj et al., 2003). Thereafter, the final phase of the feature change process is the implementation, if the feature change is to be carried out.

2.4. Research framework

A conceptual research framework is used in this study for theory building. As presented by Miles and Huberman (1994), a conceptual research framework describes in graphical or narrative form the main aspects to be studied meaning the key factors, constructs or variables, as well as the relationships among them. They explain that theory-building relies on a few general constructs that contain a set of particulars. Elements are labels of the theory which are put in intellectual “bins” including many distinct events and behaviours. Bins are formed based on theory and experience as well as from the general objectives of the study. A conceptual framework is about setting out bins, naming them, and gaining clarity about their interrelationships (Miles and Huberman, 1994).

The research framework for software product roadmapping is created based on the existing literature. The purpose of the research framework is to help to structure and present the research results of this study. The research framework for software product roadmapping is presented in Fig. 1.

The framework aims at providing answers to questions that relate to the following elements in product roadmapping: stake-

holders and process phases. The research framework also includes the perceived impacts of software product roadmapping, i.e., the main benefits and challenges faced during product roadmapping and the most critical phases of the process. The empirical results in Section 4 are presented in accordance with the research framework at hand.

It should be acknowledged, that there are also other frameworks presented in the literature that include product roadmapping, for example the reference framework for software product management presented by van de Weerd et al. (2006). However, their framework has broader perspective to the software product management than we do in our framework. Among other activities, their framework includes portfolio management, and we focus purely on the content of the product roadmapping activities and stakeholders. According to van de Weerd et al. (2006) the product roadmapping includes activities such as theme and core asset identification, and roadmap construction. These activities are included in our framework in to the product roadmapping phases. For example, theme and core asset identification are included in the capturing features phase. In our framework, the roadmap construction is not seen as separate activity of the process; instead, the product roadmap is created and revised through out the process.

3. Research design

In this section, the research methods and goals as well as research context and data collection are defined. First, research methods and goals (Section 3.1) are presented and then, the research context and data collection (Section 3.2) are described.

3.1. Research methods and goals

The research is a multiple-case study. The purpose of the multiple-case study is used either when the results from the earlier case study are verified, i.e. similar results are predicted, or when contrasting results are obtained, but for predictable reasons (Yin, 1994). This research approach was used, since the theory on the product roadmapping was not yet well formulated. Therefore, this study was started by conducting a literature review (Section 2). Its purpose was to understand the current state and to find the gaps in the literature. As a result, a conceptual framework was created (defined in Section 2.4). It contains the essential elements of product roadmapping and was used in designing the empirical part of this study.

The empirical research was carried out as a qualitative research conducted with a questionnaire-based survey study (Oppenheim, 1992) and semi-structured interviews (Järvinen, 2001). In the questionnaire-based survey method, Alreck and Settle (1995) classify three methods of data collection: personal interviewing, telephone interviewing and mail data collection, of which all were used during the research. In selecting the respondent organizations for the questionnaire survey and interviewees, a purposive sampling (Nardi, 2003) was used. Purposive sampling involves designating group of people for selection because they have some traits that are important for the study (Nardi, 2003). In this research, the questionnaire respondents were selected based on VTT's (VTT, Technical Research Centre of Finland, 2009) electronic mailing lists. Also, the questionnaire was sent to European-wide Merlin research projects' partner companies (MERLIN ITEA project 2004–2007), since they were assumed to have knowledge about the field of study. The interviewees were, then, selected based on the questionnaire respondent's experience in product roadmapping. The emphasis of the empirical research was put on the interviews. The questionnaire was used to obtain data about product roadmapping and to obtain the right persons to be interviewed.

Table 3
Empirical research design.

Research method	Research questions relating to each element		
	Stakeholders	Process phases	Critical process phases
Questionnaire	Who should participate in product roadmapping? How many persons participate in the roadmapping process in your company?	How product requirements are captured into roadmaps? How requirements are prioritised during product roadmapping?	What is the most important phase in the roadmapping process? What is the most difficult phase in the roadmapping process?
Interviews	What roles does the roadmapping process include? Who participates into the roadmapping process and to which activities in it?	How does the roadmapping process begin? What are the main phases of roadmapping process? How each product roadmapping phase is conducted?	What are the benefits of product roadmapping? What are the problems of product roadmapping?

The content of both the questionnaire and the interviews were designed according to the research framework for software product roadmapping, presented in Section 2.4. The questions were divided along the framework into three elements: stakeholders, process phases, and critical process phases. An example questions from both research methods along with the elements are presented in Table 3.

The goal of this study is to increase the current empirical evidence on product roadmapping from three elements: (1) the stakeholders of the product roadmapping (i.e. who the main stakeholders are and what kind of role they have during the product roadmapping process), (2) the product roadmapping process phases (i.e. what are the main process phases and how they are conducted), and (3) the benefits and challenges and critical points of product roadmapping. In addition, a tentative research framework for software product roadmapping is proposed (Section 2.4) based on the current literature. The research framework is for designing the research as well as ordering and presenting research results.

3.2. Research context and data collection

In this research, the empirical study was carried out in five phases: (1) formulating the questionnaire, (2) pilot testing of the questionnaire, (3) analysis of the questionnaire results, (4) interviews, and (5) analysis of findings. Fig. 2 illustrates the process of the conducted empirical study.

Firstly, a literature research was conducted in order to understand the state-of-the-art of product roadmapping. In addition, the purpose of the literature review was to focus research on a few of the most important issues from the field of research, and to prepare the proposed questionnaire based on the literature review. The data collection was carried out, firstly, in the form of a questionnaire study to collect background knowledge of the case companies on the application of product roadmapping. The interviews, then, were conducted to collect more in-depth knowledge about product roadmapping in different companies.

The questionnaire was built in a structured form to provide the respondent with specific alternatives for each question. However, in order to get more in-depth information, a respondent could also reply with an open answer to each of the questions. Thus, the questionnaire included both structured and unstructured questions. The structured questions were selected to attain a brief response time and therefore to receive more replies. Also, the structured questions

were chosen to make analysis easier and conclusions stronger. The questions for the inquiry were planned and arranged carefully in advance to attain the right form of questions and to avoid misunderstandings. Thus, the questionnaire was pre-tested inside VTT (VTT, Technical Research Centre of Finland, 2009) among researchers, and based on the pre-test results, small modifications to the questionnaire were made.

After the pilot testing of the questionnaire, in summer 2006, the questionnaire was sent to potentially interested contacts by using purposive sampling (Nardi, 2003), i.e., to companies assumed to have experience and interest in product roadmapping. An electronic mailing list was prepared and the questionnaire was e-mailed to over 600 respondents. The high receiver rate can be partly explained by the fact the mailing list was not ranked, e.g. according to respondents' roles, thus the questionnaire was also sent to those respondents that were not actually involved with the scope of research. As a result, a total of 59 responses were received, seven of which lacked experience in product roadmapping and were hence unable to fill the form. Though the resulting response rate of the questionnaire was quite low (10%), the 52 replies from 34 different companies can be claimed to provide a wide enough perspective for further analysis, especially for selection of persons to be interviewed. One of the replies, however, was excluded from the analysis due to an incompletely filled questionnaire form.

The respondent companies originated from Finland, Sweden and the Netherlands. All companies were involved in software product or service development in the field of information and communications technology (ICT) industry. Their scopes varied from own product development to the development of components for external partners. The size of the case companies, measured as the number of overall employees, was distributed among the given categories in the questionnaire (under 10 employees, 10–49 employees, 50–250 employees, and over 250 employees) with an emphasis on medium and large companies. The distribution of answers according to company size is illustrated in Fig. 3.

After the questionnaire study, the interviews were planned. The purpose was to gain in-depth knowledge on how product roadmapping was conducted in case companies. The interviews were semi-structured, and as Järvinen (2001) describes, the semi-structured interviews include structured questions and proceed along certain vital themes of the research. The vital themes of research were selected based on the literature analysis, presented

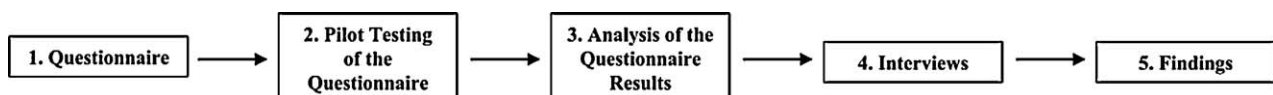


Fig. 2. Empirical research process.

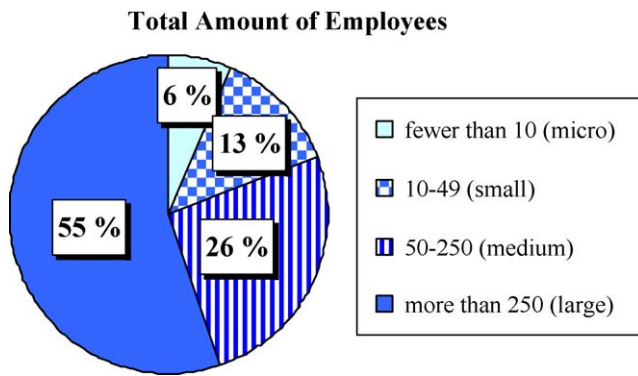


Fig. 3. Size of the case companies.

in Fig. 1, and findings of the questionnaire studies. As is characteristic of semi-structured interviews, the themes were the same for all the interviewees, but the questions varied between the different interview sessions. Equally, the interview questions were partly planned in advance, but not in a detail formed or arranged. Moreover, the intention in the interviews was to emphasize the interviewee's experiences and their own opinions on the field of study. Furthermore, the interviews could be considered focused interviews, since the respondents were interviewed personally for a short period of time, i.e. not more than an hour (Yin, 1994).

Altogether, nine persons representing eight different companies were selected to be interviewed. Two of the interviews were face-to-face, and the other seven were conducted by phone. All the interviews were recorded and transcribed. Table 4 provides a summary of the conducted interviews indicating company nationality, company size, and the role of the interviewee.

4. Results

In this section, the research evidence from the empirical study is presented and aligned with the research framework developed in Section 2.4. Firstly, in Section 4.1, the empirical evidence to identify the main identified stakeholders of the product roadmapping process is defined. In Section 4.2, the findings related to the roadmapping process and its suggested phases are presented. In Section 4.3, the perceived impact of software product roadmapping is disclosed meaning e.g. the research evidence revealing the critical phases in product roadmapping and the main benefits and challenges of product roadmapping.

4.1. Stakeholders of the product roadmapping process

In order to clarify different definitions in the literature about the stakeholders of the roadmapping process, the questionnaire respondents were asked to select different functions of the organization that should take part in the product roadmapping process. Also, the purpose was to find out the most important stakeholders.

The stakeholders were divided into the following categories: product management, finance, engineering, marketing, manufacturing, services, development, customer and partner representatives, and other. Fig. 4 illustrates the questionnaire replies relating to the stakeholders of the roadmapping process.

According to the respondents, the following were considered to be the most important stakeholders of the product roadmapping process: product management, marketing, customer and partner representatives, and development including manufacturing and engineering. From these stakeholders the customer and partner representatives were the only external stakeholders to be involved in product roadmapping based on the questionnaire results. However, all the categories received a number of responses and, thus, can be considered somehow important. Additionally, 21% of the respondents thought that other groups of representatives were also needed during the process as well. These groups included various management stakeholders, e.g., top management, senior management, and human resource management, as well as final end-users and sales personnel.

When several different functions from organization(s) participate in the roadmapping process, the number of participants can become quite large. Therefore, the questionnaire respondents were asked to describe how many persons typically participate in the roadmapping process in their company from the given categories (1–5 persons, 6–10 persons, 11–20 persons, 21–30 persons, and other). In total, 50 responses were received for this question.

It may be considered surprising that the category “1–5 persons” was selected by 34% of the respondents, even though most of the companies were considered large. Thereafter, the second highest rate was given to the category “6–10 persons” with 30% of the replies. The third largest category was “Other” with 16% of the replies, in which the respondents explained that more than thirty, fifty, or hundred persons participate in the roadmapping process. Additionally, some of the respondents described that hundreds or several hundreds, or more than two hundred persons take part in the roadmapping process. The two last categories were not very common, since the category “11–20 persons” received 14% and the category “21–30 persons” received only 6% of the total number of replies.

The questionnaire data also implicated that the company's size and the number of participants in the roadmapping process clearly correlate. According to the majority of the small companies with fewer than 50 employees, one to five persons participate in the product roadmapping process. Instead, in medium sized companies with employees from 50 to 250, the number of participants in the roadmapping process was from six to ten. Furthermore, in case of large companies with more than 250 employees the number of participants was more than 30, and could be even several hundreds, which made the process very complex, as one of the respondents explained.

The interviewees emphasized that not all the participants were necessarily present during each phase of the process. Instead, the participants only participated in those phases that affected their

Table 4
Summary of interviewee profiles.

Interviewee	Company nationality	Company	Company size (employees)	Role of the interviewee
1	Finnish	A	>250	Manager
2	Finnish	A	>250	Group manager
3	Finnish	B	50–250	Chief technology officer (CTO)
4	Finnish	C	>250	Product planner
5	Finnish	D	>10	Program director
6	Finnish	E	>250	Group manager
7	Finnish	F	>250	Chief engineer
8	Swedish	G	>250	Senior researcher
9	Finnish	H	50–250	Manager

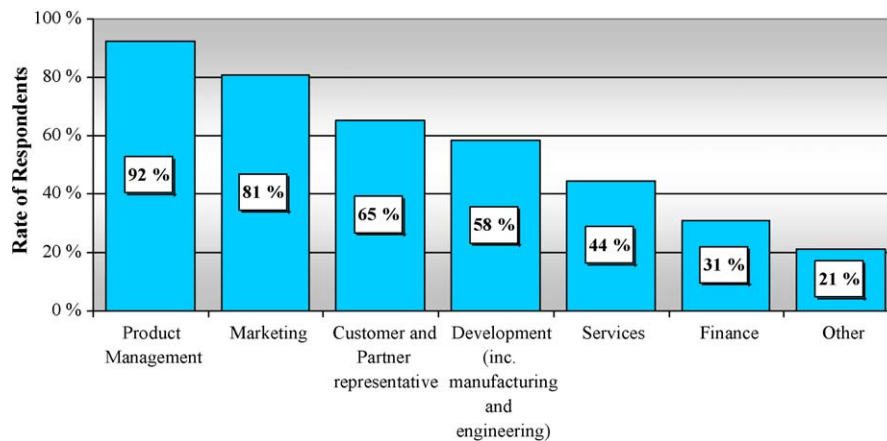


Fig. 4. Stakeholders of the product roadmapping process.

work or knowledge. Especially, when the company was large, there were different types of roles from different kinds of companies that were involved in different phases of roadmapping. Thus, different meetings were reported to be held with different groups of participants to focus on specific areas of the product. One of the interviewees pointed out:

“Separate feedback or prioritisation sessions should be held, and joint meeting are held to discuss all issues.” [Manager, company H]

Instead, in smaller companies when the process included fewer participants, all the participants could be present in each phase and meeting of the process.

According to the interviews, the product roadmapping process participants had only two basic roles: as a member of the product roadmapping team and as a product or solution owner. In addition, there was a third role mentioned in the literature, the facilitator's role, which was not considered to be important by the interviewees. According to the interviews, the most important role was considered the owner's role as this person had the overall idea of the desired roadmap, and there should be someone in possession of the roadmap. The role of the owner was also considered to include the collection of input, holding the roadmap together, making the needed changes to the roadmaps, and taking care of the information flow both within and outside the company. Instead, the stakeholders of the roadmapping team were to bring input from different viewpoints to the roadmap, e.g. to schedules and to product features. The team was also responsible for evaluating and prioritising features as well as reviewing created roadmaps.

4.2. Product roadmapping process

Among the interviews, product roadmapping was identified as a continuous process where the roadmapping team has meetings biweekly, quarterly, or biannually, in which roadmaps are updated and reviewed. The majority of the interviewees replied that the product roadmapping process begins with customer requirements. These requirements could be, for instance, proposals for improvement or new product features as well as the customer's goals or expectations. The requirements could also come from the company's internal research unit or through competitor analysis. In some cases, changes in the standards were reported to cause the trigger for a product roadmapping. The product roadmapping process could also begin with:

“By defining what product features were emphasised in the markets.” [Group Manager, company A]

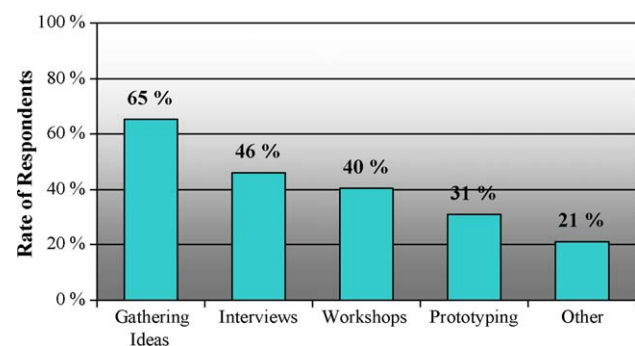


Fig. 5. Methods for capturing features.

The product roadmapping process consisted of three to six phases according to the interviews. Although the number of phases varied, in almost all the companies the same tasks could still be identified as elements of the product roadmapping process.

4.2.1. Capturing features

As mentioned in the literature, there are several methods, which can be used during capturing features. To find out what kind of feature capturing methods were used in the case companies, the questionnaire respondents were asked to select the used method(s) from the given categories (with prototyping, with interviewing, gathering ideas over time, in some kind of workshops, and other). All 52 responses were included in the analysis; most of the respondents selected more than one alternative.

Although the literature in the field of the study recommends using workshops for capturing features, the questionnaire replies revealed that some other procedures were used more often than workshops. Accordingly, the most commonly used method was gathering ideas over time with 65% of the replies and after that, interviews with 46% of the replies. Only 40% of the respondents used workshops to gather features, and 31% of the respondents used prototyping. Additionally, 21% of the respondents used some other methods to capture features. These methods included, for instance, market and technical research, following market development and standards, as well as analysing the distribution channels and their requirements. The most commonly used methods for capturing features are presented in Fig. 5.

According to the interviews, during capturing features, the features originated from several sources of information such as from current market trends and standards. Thus, knowledge of different participants was combined. The stakeholders of capturing features were typically the collaborators, sales, management, or product

architects, as well as from a company's own research, for example, through customer and competitor analysis. In addition, one of the interviewee's replied:

"Features are typically captured and feedback is collected through trade shows and by distributing a free evaluation version through the internet". [CTO, company B]

In some case companies, the idea of the product was created together with collaboration partners, e.g. through brainstorming. In the other case companies, possible participants and co-operation partners were clarified after a product idea was created inside the company. However, as one of the interviewees noted:

"Capturing features has two phases: someone has an idea for a product feature and another one writes down what the feature actually means. Based on that information it can be verified that everyone has understood the idea of the product feature correctly." [Chief Engineer, company F]

4.2.2. Analysing features

Various methods were used for analysing features in the case companies based on the interviews. Some of the case companies used domain-specific knowledge and experience as well as interviewed experts during the analysis phase. Further, in one case company, the major features (i.e. features requiring more effort to be implemented) were analysed through a feasibility study, as suggested in the literature. The feasibility study included both a technical and cost analysis. Based on the analysis estimations for the revenues, sales and implementation could be made. Minor features were not analysed, but rather were further defined in order to make the decision whether they fit into the product content or not. In the latter case, they were dropped or postponed as candidates for the next product release. Usually, the analysis of features was conducted by the experts from different stakeholder groups, e.g. from the viewpoint of sales, technical or strategic value. However, in one case company:

"Analysing features is the responsibility solely of the product manager or person in charge of the product." [Product Planner, company C]

Typically, the three most important factors in analysing features were identified as: (1) the estimated cost, (2) technical requirements (e.g., specific hardware needs), and (3) the central use cases (e.g. when and how the functionality is being used, and what else should function at the same time).

In more detail, analysing features was found to consist of three phases. Firstly, it was verified whether the features were understood correctly, for example, whether the features were recorded exactly, and if the feature description included information concerning the feature's functionality and limitations. Secondly, it was figured out what kind of technology would support the implementation of the features. This was typically done by the architects and technical persons and could involve prototyping or sketching the software architecture. This was reported to help in understanding the implementation of the features from a technical viewpoint. Thirdly, after creating basic knowledge and understanding, it was estimated how much it would cost in terms of time, money, and external resources to implement the features.

4.2.3. Prioritising features

According to the literature, the product features can be prioritised both with formal or informal methods. To find out what kind of prioritisation methods were typically used, the questionnaire respondents were asked to select either formal, informal or both methods. The clear majority of the 52 respondents selected

informal methods with 69% of the total amount of replies. Thereafter, 23% of the respondents used formal methods only, and 8% of the respondents used both methods for prioritising features. The formal prioritisation methods included Analytic Hierarchy Process (AHP) (Saaty, 1986), Quality Function Deployment (QFD) (Griffin and Hauser, 1993), EVOLVE (Greer and Ruhe, 2004), and Distributed Prioritisation (Regnell et al., 2001). Additionally, in one case company, formal methods contained business cases, probability analysis (win/lose), and customer and market importance balancing.

Moreover, two of the interviewees had experience on the use of formal prioritisation methods; Distributed Prioritisation and AHP. Distributed Prioritisation was used:

"...when it was supported with tools allowing simulation of multiple approaches and weighting of answers." [Manager, company H]

It was also used to support distributed knowledge, since within a global company it was important to gather distributed priorities from all perspectives. Instead, AHP was used to find out focal points, for instance. Accordingly, also AHP was reported to support distribution. That was partly because the case company's user interface supported sharing and distributing information between companies.

According to the interviews, both functional and non-functional requirements were prioritised at the same time. In addition, there were no special methods for prioritising non-functional requirements. Prioritisation was typically informal, and the informal prioritising methods varied among the case companies. For instance, the priorities could be created by using a calculation system. In that case, each feature was given a point, which could also be a weighting factor, work contribution, or incurring costs. After giving the points, it could be seen which of the features had won. Based on the points, the order of priority was formed. Instead, in some case companies, the number of customerships per feature or values related to technical importance, market value, or ease with return on investment (ROI), guided prioritisation. On the other hand, in some cases, more information relating to the features was needed to be considered during the prioritisation. For example, customer preferences, legislative or release specific features, and real world matters guided prioritisation.

Typically, the order of priority of the features was a result of a collective decision in the roadmapping team. In some cases, the decision could be made also by the product manager, CTO, the owner, or the person in charge of the product. Usually, minor problems concerning the priorities were handled inside the roadmapping team and the major problems were resolved by the management team of the company or companies. Additionally, in some case companies, the customer made the final decision concerning the priorities, especially, when intermediate versions from the product were important to the customer.

4.2.4. Roadmap validation and agreement

Commonly, there were two means for performing validation according to the interviews. The roadmaps were either validated through reviews or through the unit's business improvement and customer feedback.

According to the first group of interviewees, roadmaps were validated through negotiations, meetings, or reviews. Accordingly, reviews were considered the most efficient way of performing validation, especially when roadmaps were reviewed with an adequate number of persons. During the review process, inputs as well as comments were collected, and if there was any essential new information, the necessary changes to the roadmaps were made. Hence,

in order to widely spread and review the roadmaps:

“Roadmaps are kept in either paper or electronic format.” [Program Director, company D]

Additionally, the roadmap validation could be manifested in contracts. It could even be a legal contract, if financial matters were involved. Moreover, when roadmaps were created together with collaborators, validation of the roadmap was also performed in mutual meetings with the partner. Typically, the customer confirmed that the roadmap was good.

Instead, according to the other group of interviewees, validation took place when the customers started to buy or not to buy the product. Thus, validation came through the unit's business improvement and customer feedback. If negative results were obtained they were analysed, for instance to reveal why product development had gone in an unprofitable direction or why the newest version did not answer to the customer's needs. Roadmap validation could also come through commercial success, e.g. as a function of the number of sold products.

The roadmap agreement was often made in a meeting together with the roadmapping team. The agreement could also be made by the product owners, the product managers, or CTO, depending on the case company, and if needed, the management participated in the meetings as well. In such a meeting, the product roadmapping team was reported to make a mutual decision about the roadmap from which point onwards the development work would then proceed. The roadmap agreement was made in order to have a mutual understanding about the product that was being developed. The agreement also made the roadmap official. Thus, with the agreement, it could be ensured that commitment existed and everybody knew the decided matters.

4.2.5. Change management of the roadmap

According to the interviews, changes to the roadmaps came from delays in the product implementation as well as when new or unnecessary product features were discovered during product development. Additionally, when partners or customers were informed about the new solutions, they typically brought up matters that might not have been taken into account earlier. These new matters had to be either brought out with the ongoing releases or left at the roadmap stage for forthcoming product releases. The case companies reported having meeting practices for change management, in which change requests were handled and decisions concerning the change were made. One of the interviewees highlighted:

“We try to manage changes that affect schedules or money in joint meeting, from which meeting memos are created.” [Chief Engineer, company F]

Thus, it was found important to record the meetings, e.g. create notes or meeting memos, in case there would be a need to verify the change decisions later on.

The roadmap change process typically went as follows. First, either one of the collaboration partners noticed the change request. Secondly, the effects of the change were analysed, i.e. an impact analysis was conducted. Third, changes were approved together with the partners, or a customer was requested to approve the changes. That was because all the decisions had to be conscious and those that were mostly affected by the change had to be able to affect the change decisions. Thereafter, it was verified that everyone had understood the changes. Finally, the changes were added to the rest of the features, i.e., a new roadmap or updated version of a roadmap was created.

The decision-maker of the change was reported to depend on the importance of the matter to be changed. Minor changes could

be made by the product manager or CTO. Major changes were managed by the roadmapping team. Major changes were largely related to schedules, deleting important or key features from the roadmap, or adding bigger features to the roadmap, etc. Thus, these changes had to be communicated and approved by the same forum that had approved the roadmap. Instead, when the customer was financing the product, then the final decision was made by the customer. Moreover, in other collaboration situations, the first, preliminary decision concerning the change was made inside the company. Thereafter, the change decision was negotiated together with the collaboration partners in meetings, and they tried to reach an agreement. When unsolvable problems occurred during meetings, then management of the companies made the final decisions concerning the change. Therefore, in collaboration, the changes could not be unilaterally decided, instead they had to be approved mutually. Usually, the rules concerning the change management were known and approved before cooperation, for example, in the cooperation agreement.

4.3. Perceived impact of software product roadmapping

The interviewees had convergent ideas about product roadmapping. Based on the interviews, a roadmap is a plan about the company's future directions; in other words, it is a leading map where the company is going with its products. Thus, it is the means of structuring and arranging the product development, in order to know how to use certain resources. A roadmap holds the product development together by guiding what is to be done and when. Hence, product development is somehow deterministic and enables steering of the product implementation. A roadmap also gives a clear focus in the product development, and provides high-level understanding of scoping the strategy. On the other hand, one of the interviewees pointed out:

“Clear strategy allows better planning and commitment to the set plans.” [Manager, company H]

Product roadmapping also improves predictability, and reduces surprises during product development. With a roadmap, tasks to be done can be prioritised, and thus resources can be allocated to the most profitable projects. It can even be verified from the roadmap that the right things are done at the right time. Additionally, with a good roadmap the customers' needs can be met with a product that they really want. Hence, roadmaps offer a competitive advantage. Moreover, a roadmap is a central tool for communication, and therefore it should be shown to the company's own staff and to partners. It gives a clear idea what is about to be done and enables communication about forthcoming strategic projects.

4.3.1. Importance of the phases

The questionnaire respondents were asked to rank the roadmapping phases from the most important to the least important (1 = most important and 5 = least important). The phases to be ranked were divided into the following categories: capturing features into roadmaps, analysing features, prioritising features, roadmap validation and agreement, change management of the roadmap, and other. Two of the replies were left outside the analysis since several “most important” alternatives were selected. Hence, the total number of replies was 50. It should be noticed that not all of the respondents ranked all the categories. Instead, some of the respondents only selected one to three phases that were considered the most important, and left the other phases outside the ranking. Thus, some of the categories have fewer replies. Fig. 6 illustrates the phases of the product roadmapping process and the ranking order given by the respondents.

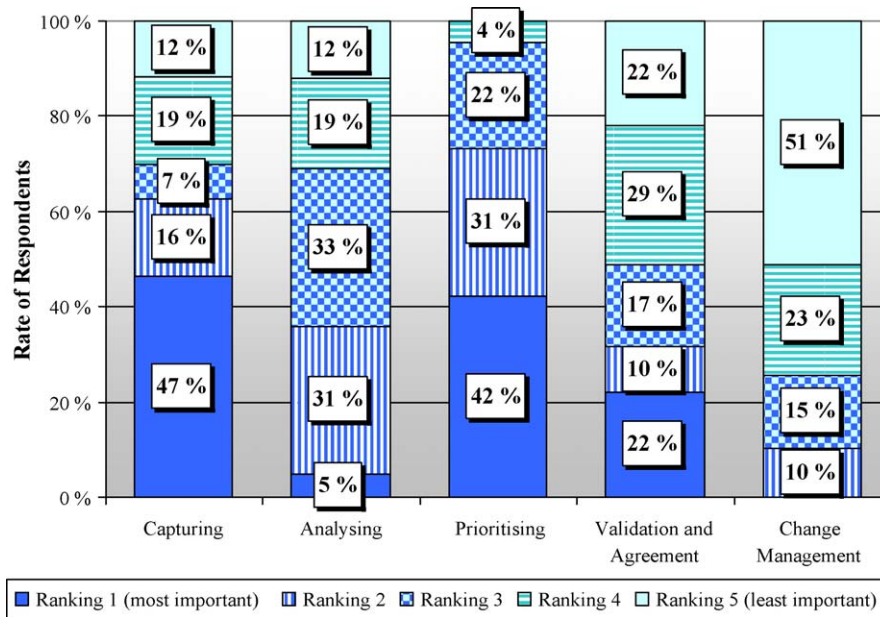


Fig. 6. The most important phases in product roadmapping.

The majority of the most important replies, 47% altogether, were given to the capturing features phase. After this came the prioritising features phase with 42% of the most important replies. Since none of the respondents thought that prioritising features was the least important phase in roadmapping, the prioritising product features was considered the most important phase in the product roadmapping, and then the capturing features. The roadmap validation and agreement was considered the third most important, and the analysing features phase was the fourth most important. The change management of the roadmap was not considered so important, since none of the respondents ranked this phase as the most important. Instead, the majority of replies gave this phase the rating of least important. This can be explained by the fact that the change management will not help, if all the other phases have gone wrong.

4.3.2. Difficulties of the phases

Thereafter, the respondents were asked to select one of the product roadmapping phases that was the most difficult. According to the clear majority of replies, the most difficult phase was prioritising features with 46% of the replies. The other phases were distributed quite equally among the replies; capturing features with 19% of the replies, validation and agreement with 17% of the replies, analysing features with 10% of the replies, and change management of the roadmap with 8% of the replies.

In order to find whether the company's size affects the most difficult phase in roadmapping, the companies were divided into the three size groups (small: fewer than 49 employees, medium: 50–250 employees, and large: over 250 employees). Based on the replies, both the medium and the large sized company groups thought that the most difficult phase of roadmapping was the prioritising features phase. Instead, according to small companies, the most difficult phase in the product roadmapping was the roadmap validation and agreement phase. Additionally, large companies faced difficulties during all the phases of roadmapping unlike the small and medium sized companies that faced difficulties only during prioritising features, roadmap validation and agreement, and capturing features into product roadmaps.

Furthermore, it was analysed whether the lifetime of the company's products had an effect on the most difficult phases of the roadmapping. The product service lives in use by customers were

divided into the following categories: a lifetime from one to six years, a lifetime over six years, and another that included products with varying lifespans from one to more than 10 years. The companies with a product lifetime between one and six years had difficulties during all the phases of roadmapping except during the change management phase. Additionally, the companies with a product lifetime over six years did not face difficulties during the roadmap validation and agreement. Instead, companies that selected the category "Other" faced only difficulties during the roadmap validation and agreement, and the capturing features phase.

According to the interviews, there are problems relating to almost every phase of the product roadmapping process. Most commonly, the problems seem to relate to prioritising the features, managing changes, and maintaining roadmaps. Also, sharing information, communication, and making the roadmap agreement were considered difficult.

Collecting input was considered difficult, and more precisely, getting the right information and accurate knowledge was problematic. Also, background research and finding out both competitors' and customers' opinions was difficult. Hence, uncertain predictions about the future were attempted to be avoided by collecting feedback on the prediction accuracy. Additionally, it was highlighted that at the beginning of the product, the roadmaps could be more accurate which meant that a more precise product design could be created right from the beginning of the process.

In prioritisation, problems were caused by uncertainty about which features should be taken into which product version. It was also described that the strongest opinion may not offer the best understanding of the values, and thus prioritisation should be done carefully. Relating to prioritisation one interviewee revealed:

"If not done with clear vision and with strategic thinking it will lead to changes in plans and results in frustration very ineffective product release cycles." [Manager, company H]

Also, prioritising customer requirements was considered a continuous problem and a challenge in the roadmapping process. Making an agreement was also considered difficult, since combining and processing different viewpoints was complex. Especially, consolidation of different wishes, needs, and technical implementation possibilities in an economical frame was considered

hard. Moreover, managing and maintaining product roadmaps was thought difficult. The problems were caused by a large number of changing matters, and the more precise the roadmaps were the harder they were to maintain.

Communication about features across organizational borders to ensure that everyone involved understands the meaning of the features was considered difficult. Additionally, communication in large companies was considered problematic. Thus, it was suggested that sharing information and its visibility should be improved. Moreover, it was feared that information would get into the wrong hands, since it was problematic to verify persons who should know about the roadmap, and that they were well aware of the content of the roadmap. Thus, it was a matter of concern, how to get the information without the danger of data leakage. Hence, the purpose was to prevent the information from the roadmap from being passed on to competitors.

5. Discussion

In this section, the research results are discussed. The literature review findings are compared against the empirical research results. In Section 5.1, the empirical findings from the product roadmapping stakeholders are compared against the literature findings presented in Table 1 and, in Section 5.2, the empirical findings from the product roadmapping process are compared against the literature findings presented in Table 2. Also, at the end of this section, common practices in product roadmapping are summarized. In Section 5.3, the critical process phases of product roadmapping are discussed.

5.1. Product roadmapping stakeholders

Based on the empirical research findings, the product roadmapping process only seemed to involve two main actors: the roadmap owner and members of the roadmapping team. Empirical evidence did not support the use of a facilitator, as proposed in the literature (e.g. Phaal et al., 2003a; Wells et al., 2004). Furthermore, in the literature (e.g. McCarthy, 2003), it is suggested that the roadmapping team is, among others, responsible for creating the roadmaps. However, the research results indicate that the roadmaps are created by the owner, while the main task of the team is to give input to the roadmap.

As presented in Table 1 several different stakeholders were mentioned in the literature to participate in the roadmapping process, the empirical findings exposed the following main actors: product management, marketing, customer and partner representatives, and development including manufacturing and engineering. Therefore, for example, participants from finance were not seen as important as suggested in the literature (Phaal et al., 2000a,b, e.g. McCarthy, 2003; Phaal et al., 2003a). The customer and partner representatives were the only external stakeholders to be involved in product roadmapping. Instead, according to van de Weerd et al. (2010) the external stakeholders include representatives from the market, partners, and customers. Additionally, based on the empirical findings, the number of participants in the roadmapping process was directly connected to the size of the company. In larger companies, more stakeholders participate in the process. However, not all the participants were necessarily present at each phase of the process. Instead, they only participated in the phases that affected their work or where their particular knowledge was needed or relevant.

It was suggested in the literature (presented in Table 2), that the roadmapping process is different in every company (Phaal et al., 2004; Groenvelde, 2007). However, the empirical evidence suggests that the contents of the roadmapping processes are still almost the same. The tasks to be done during the roadmapping process were

the same in spite of the fact that the case companies had identified a different number of phases and different names for the phases. As a result, the following phases of a common product roadmapping process are proposed in this paper, i.e., capturing features, analysing features, prioritising features, roadmap validation and agreement, and change management of the roadmap.

5.2. Product roadmapping process

According to the empirical findings, the product roadmapping process was considered a continuous process that was a part of the product development process. Unlike what is suggested in the literature (Vähäniitty et al., 2002) that roadmapping process starts with defining or revising, and then analysing the strategic mission and vision of the company, the results of this research propose that the product roadmapping process seems to begin with the customer requirements. In some cases, the process also begins from product requirements emerging from the company's internal research, or through competitor, market or standard analysis.

It was presented in the literature (e.g. Phaal et al., 2003a) that workshops were most commonly used for capturing features. Instead, the questionnaire study revealed that gathering ideas over time was the most commonly used method for capturing features and interviews after that. Furthermore, the features could come from several sources of information, e.g. market trends and standards. Combining knowledge from different participants was seen important, and therefore several stakeholders were advised to be involved in capturing features, e.g. collaborators, sales, management, product architects, and market researchers.

The methods for analysing features were almost alike in the literature (e.g. Fickas and Nagarajan, 1988; Zhang et al., 2005) and the empirical results. For example, the features were analysed by using domain-specific knowledge and experience as well as interviewing experts. However, during the interviews it was noted that there is a difference between analysing the major and minor features. The major features were analysed by using such methods as mentioned in the literature, e.g. feasibility study. Instead, the minor features were not analysed; on the contrary, they were only planned and then considered whether they would fit into the product's content or not. Furthermore, the empirical evidence revealed that analysing features includes following important factors to be noticed: (1) verify understanding, (2) clarify implementing technology, and (3) create cost, time, and work estimations. Analysing features was typically conducted by the experts from different stakeholder groups, e.g. sales and people with technical or strategic perspective.

According to the literature, the product features could be prioritised with formal (e.g. McCarthy, 2003; Phaal et al., 2003b; Groenvelde, 2007) and informal (e.g. Phaal et al., 2003a; Blotner, 2004) prioritisation methods. Empirical evidence suggests that features are typically prioritised by using informal prioritisation methods and both functional and non-functional requirements were prioritised at the same time. However, the methods used for informal prioritising differed from the ones proposed in literature. For instance, Blotner (2004) suggests using identity information for initial feature prioritisation in which each feature is presented and input from all team members is collected. According to the interviews, the priorities could be created by using a calculation system; or via a number of customerships per feature or values related to technical importance, market value, or ease with return on investment (ROI), guided prioritisation. Based on the interviews, more information relating to the features was considered during the prioritisation. For example, customer preferences, legislatively or release specific features, and real world matters guided prioritisation. As suggested in the literature (Phaal et al., 2003a), the roadmapping team prioritised the features. However, it was also pointed out in the interviews that the decision could also be made

Table 5
Essential factors influencing product roadmapping.

Phase	Essential factors
Capturing features	Gathering ideas over time and performing interviews to capture features Features come from several sources of information, e.g. market trends and standards Stakeholders include, e.g. collaborators, sales, management, product architects, and market researchers
Analysing features	Using domain-specific knowledge and experience, and interviewing experts Analysing and conducting feasibility study for major features Making a decision whether minor features fit into the product content or not Important factors: (1) verify understanding, (2) clarify implementing technology, and (3) create cost, time, and work estimations Stakeholders include, e.g. a sales representative, and people with technical or strategic perspective
Prioritising features	Using informal prioritisation methods, e.g. a calculation system Prioritising functional and non-functional requirements at the same time Several factors guide prioritisation, e.g. customer preferences, legislative and/or release specific features Stakeholders include, e.g. roadmapping team, product manager, CTO or the person in charge of the product
Roadmap validation and agreement	Validating roadmaps: (1) through negotiations, meetings or reviews to collect input and comments, or (2) through unit's business improvement, customer feedback or product's commercial success Making a roadmap agreement: in a meeting to have a mutual understanding about the product and to make the roadmap official Stakeholders include, e.g. roadmapping team, product owner, product manager, or CTO
Change management of the roadmap	Changes come from delays in product implementation or discovering new or unnecessary product features during product development Using meeting practices to handle change requests and making decisions concerning the change Roadmap's change management process: (1) change request, (2) impact analysis, (3) approve changes and verify understanding, and (4) revise roadmap Stakeholders of a minor change: product manager or CTO Stakeholders of a major change: roadmapping team

by the product manager, CTO or the person in charge of the product. Additionally based on interviews, if consensus could not be reached in the team, the management team of the company or companies made the final decision.

According to both literature (e.g. McCarthy, 2003) and empirical findings, the roadmaps were typically validated and agreed on in meetings. However, according to some of the case companies, the validation occurred when the customers started to buy or not to buy the product. Hence, the validation could also be based on the unit's improvement, customer feedback, or a product's commercial success. The empirical evidence also revealed that the roadmap agreement was made in a meeting together with the roadmapping team to have a mutual understanding about the product and to make the roadmap official. The roadmap agreement could also be approved by the product owner, product manager, or CTO.

Empirical evidence pointed out that the changes to the product roadmaps come from delays in product implementation or when new or unnecessary product features were discovered during product development. It was suggested to use meeting practices to handle change requests and make decisions concerning the change. Moreover, based on the empirical findings, the change management process of the product roadmap consisted of four phases instead of the six phases as proposed in the current literature (Pozgaj et al., 2003). These phases were: (1) change request, (2) analysis of the changes' effects, i.e. impact analysis, (3) approving changes, and (4) creating a new roadmap or revising a roadmap. Additionally, the findings revealed that in the product roadmapping process there was no official CCB during the change management phase as suggested in the literature (Pozgaj et al., 2003). Instead, the decisions concerning the change were made based on the importance of the matter to be changed. Minor changes were made by the product manager or CTO, while major changes were managed by the roadmapping team.

The findings of the empirical research relating to common practices in product roadmapping are summarized in Table 5. The table presents essential factors influencing the different phases of the roadmapping process.

5.3. Critical process phases

The empirical findings indicate that the most important phase of product roadmapping is prioritising features and then the capturing features. The most problematic areas of product roadmapping are prioritising features, managing changes and maintaining roadmaps as well as sharing information, communication, and making a roadmap agreement. The problems in prioritising features were caused by uncertainty about which features should be included in the product or which features should be excluded or left for the forthcoming product releases. To solve this problem, the empirical evidence revealed that prioritisation should be done carefully, with a clear vision in mind and with strategic thinking. Mintzberg (1994) also highlights the importance of strategic thinking by stating that the most successful strategies are visions, not plans. Thus, the difference between planning and strategic thinking should be noticed in order to be productive. The interviewees also suggested some informal prioritisation methods to solve difficulties in prioritising features, e.g. a calculation system. Reaching an agreement was considered difficult, since combining and processing different viewpoints was hard. Meeting practices were suggested to solve this problem. Managing and maintaining roadmaps was though troublesome because of a large number of changing matters. It was suggested that more time should be used in the first phases of product roadmapping (i.e. capturing and analysing features) to reduce changing matters at the end. Also, it was noticed that the more precise the product roadmaps were the harder they were to maintain. Thus, the product roadmaps should only contain the needed information and nothing extra.

6. Conclusions

This article provided an empirical understanding and offered experiences about product roadmapping, which have not been thoroughly explored previously. The main goals of the study were to increase the current empirical evidence on product roadmapping by defining the main stakeholders and their roles during the pro-

cess, organizing the product roadmapping process, establishing the main benefits and challenges faced during the process, and identifying the most critical phases of the process. A research framework for software product roadmapping was created based on the existing literature in order to help to structure and present the research results. The framework provides answers to the research questions and relates to the following elements in product roadmapping: stakeholders, process phases, and the perceived impact of product roadmapping. The research findings increase the product roadmapping knowledge and help companies to develop their own product roadmapping processes. For example, the study identifies the essential factors influencing product roadmapping (Table 5). The findings of the study also help the companies to focus research and development work on the most critical parts of the product roadmapping process, i.e., prioritising and capturing features as well as the change management of the product roadmap. From scientific perspective, the study aims at increasing the body of knowledge in the area of product roadmapping while also identifying future research opportunities in the field of software product roadmapping.

Results of the research indicate that a roadmap is a plan about the company's future directions. It is a leading map where the company is going with its products. Thus, it is the means of structuring and arranging the product development, in order to know how to use certain resources. A roadmap holds the product development together by guiding what is to be done and when. Hence, product development is somehow deterministic and enables steering of the product implementation. Furthermore, product roadmapping is a continuous process, since the roadmapping team has meetings regularly (e.g. biweekly, quarterly, or biannually) in order to create, update or review roadmaps. The product roadmapping team consists of several stakeholders, e.g. at least the following stakeholders are seen as important in the process: product management, marketing, customer and partner representatives, and development. Also, based on the empirical findings, feature management is seen as the key aspect in product roadmapping and accordingly the product roadmapping process is proposed to consist of the following phases: capturing features, analysing features, prioritising features, roadmap validation and agreement, and change management of the roadmap. The most critical phase of product roadmapping is prioritising features. The most problematic areas of product roadmapping are prioritising features, managing changes and maintaining roadmaps as well as sharing information, communication, and making a roadmap agreement.

The research results apply to companies that are involved in software product or service development in the field of ICT industry. In this research, it has not been analysed how, for instance, the products and procedures of the case companies differ from each other and if these have effect on the interviewees' opinions or cause some external validity threats for the study. However, all the companies of the study were selected from ICT sector to provide diverse views from the same field. For example, the analysis criteria for analysing features of different types of software products can vary as well as the methods for prioritising features. Also, it should be noticed that the research framework for software product roadmapping is created based on the existing literature at the time of research. Thus, the framework could have been different if it would have been created now, for instance. The limitations of the study also relate to the fact that most of the case companies were large companies with more than 250 employees, thus the results may be applied only to an appropriate extent to smaller companies. Hence, limitations of the study suggest directions for future research, for example the product roadmapping research could be extended to small, and medium sized companies. Additionally, it should be noticed that both the interviewees and the questionnaire respondents have subjective perspectives to the issues asked, thus

the answers reveal only the replier's opinions and not the whole opinion of the company.

The product roadmapping research is going to be continued by combining the product roadmapping and agile development methods. That is because a new type of agility is seen important in order to survive in the turbulent and competitive software business environment. Furthermore, as in this research, we have identified that prioritising features and capturing features are the most critical phases of product roadmapping they could be investigated in a more detailed way in the light of agility as well as global software development in the future. Also, through the questionnaire study, topics for the future research were found, for example why and what are the reasons that the change management of the roadmap was considered the least important phase in product roadmapping by the respondents. In this research, this was out of the scope because the focus of the research was set differently, i.e. interviews focused on the most important and problematic areas of the product roadmapping. Also, the aim is to research factors relating to the product roadmapping process itself in order to find out if the process of creating and updating product roadmaps is as straightforward as is typically suggested. For example, McCarthy et al. (2006) have proposed new product development to be regarded as a complex adaptive system (CAS) of decisions. According to them, new product development is seen as more complex than most of the models present it. The intention of further research is also to get more in-depth knowledge of product roadmapping in industry by observation, for instance.

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